```
clear; clc; close all;
run(fullfile(fileparts(fileparts(mfilename('fullpath'))),
'setup_project.m'));

s = graph_stats(make_star(5))
s = graph_stats(make_ring(5,1))
s = graph_stats(make_complete(5))
s = graph_stats(blkdiag(make_complete(3), make_complete(3)))

W = [1 1; 0 1];
x0 = [0; 1];
K = 3;
X = simulate_linear(W, x0, K)
disp(norm(X(:,end) - W^K * x0))

n  = 150;
nL = floor(n/2);
nR = n - nL;
rng(105); P0 = [5*rand(nL,2); 5*rand(nR,2) + [6 0]];

rmin = NaN;
for j = 1:200
    r = j/10;
    if isfinite(graph_stats(make_disk(P0, r)).diameter)
        rmin = r;
        break
    end
end
fprintf('r_min = %.1f\n', rmin);

G4 = make_disk(P0, rmin);
s4 = graph_stats(G4);

kmax = floor((n-1)/2);
ringEdges = zeros(kmax,1);
for k = 1:kmax
    ringEdges(k) = graph_stats(make_ring(n,k)).edges;
end
[~, kbest] = min(abs(ringEdges - s4.edges));
fprintf('T4 edges = %d, ring k = %d (ring edges = %d)\n', ...
        s4.edges, kbest, ringEdges(kbest));

Gs = {make_complete(n), make_star(n), make_ring(n,kbest), G4};
names = {'T1 Complete'; 'T2 Star'; 'T3 Ring'; 'T4 Disk'};

Edges = zeros(4,1); MeanDeg = zeros(4,1); MaxDeg = zeros(4,1); Diam =
zeros(4,1);
for i = 1:4
    st = graph_stats(Gs{i});
    Edges(i)   = st.edges;
    MeanDeg(i) = mean(st.degree);
    MaxDeg(i)  = max(st.degree);
```

```
    Diam(i)    = st.diameter;
end
T = table(Edges, MeanDeg, MaxDeg, Diam, 'RowNames', names);
disp(T);

figure;
plot(graph(G4), 'XData', P0(:,1), 'YData', P0(:,2), ...
     'NodeColor', 'k', 'EdgeColor', [0.6 0.6 0.9], 'MarkerSize', 3);
axis equal; grid on; xlabel('x'); ylabel('y');
title(sprintf('T4 disk network, r = %.1f', rmin));

G1 = Gs{1}; G2 = Gs{2}; G3 = Gs{3};
save(fullfile(projectRoot, 'Part1', 'part1_results.mat'), ...
     'P0', 'rmin', 'kbest', 'G1', 'G2', 'G3', 'G4');
```

```
s =

  struct with fields:

      degree: [5×1 double]
       edges: 4
    diameter: 2


s =

  struct with fields:

      degree: [5×1 double]
       edges: 5
    diameter: 2


s =

  struct with fields:

      degree: [5×1 double]
       edges: 10
    diameter: 1


s =

  struct with fields:

      degree: [6×1 double]
       edges: 6
    diameter: Inf

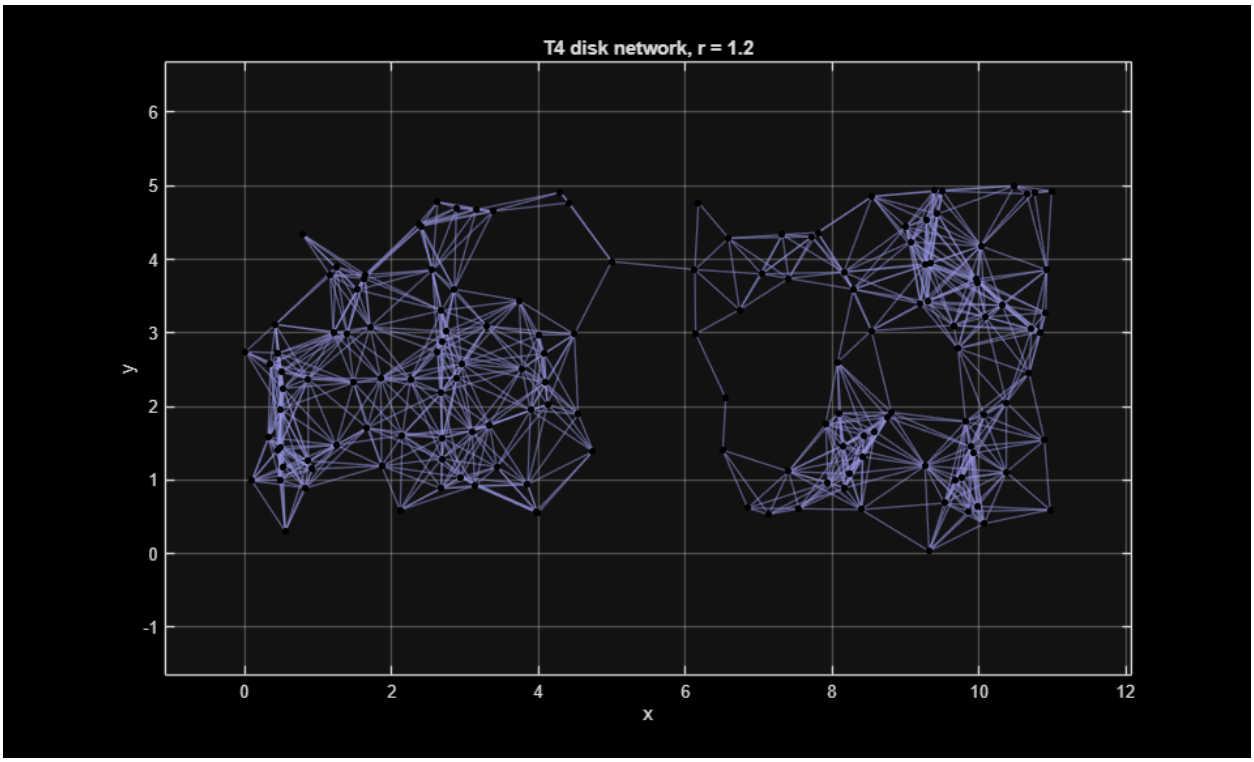

X =
```

```
     0     1     2     3
     1     1     1     1

     0
```

```
r_min = 1.2
T4 edges = 874, ring k = 6 (ring edges = 900)
                    Edges    MeanDeg    MaxDeg    Diam
                    _____    _______    ______    ____

    T1 Complete     11175        149       149       1
    T2 Star           149     1.9867       149       2
    T3 Ring           900         12        12      13
    T4 Disk           874     11.653        21      14
```



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